

Physical Properties

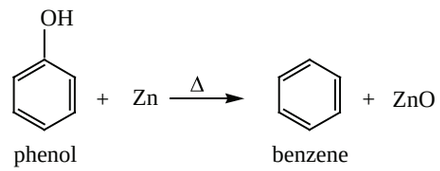
Phenol is colourless, needle shaped crystal, hygroscopic, corrosive and poisonous. It turns pink on exposure to air and light. The simplest phenols are liquids or low melting solids, they have quite high boiling points. Phenol is slightly soluble in water because of hydrogen bonding. However other substituted phenols are essentially insoluble in water.

Chemical Properties:

Reactions involving -OH group.

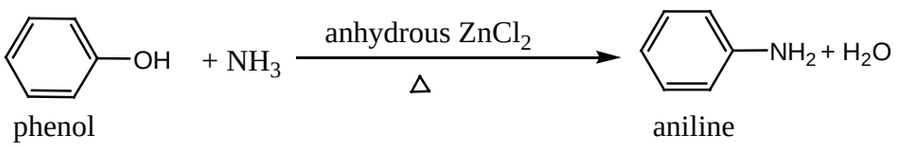
a) Reaction with Zn dust:

Phenol is converted to benzene on heating with zinc dust. In this reaction the hydroxyl group which is attached to the aromatic ring is eliminated.



b) Reaction with ammonia:

Phenol on heating with ammonia in presence of anhydrous $ZnCl_2$ gives aniline.



c) Formation of esters:

Schotten-Baumann reaction :

Phenol on treatment with acid chlorides gives esters. The acetylation and benzoylation of phenol are called Schotten-Baumann reaction.

$$\underset{\text{phenol}}{C_6H_5OH} + \underset{\text{acetyl chloride}}{CH_3COCl} \xrightarrow[Py]{NaOH} \underset{\text{phenyl ethanoate}}{CH_3\text{-}COOC_6H_5} + HCl$$

$$\underset{\text{phenol}}{C_6H_5OH} + \underset{\text{benzoyl chloride}}{C_6H_5COCl} \xrightarrow[Py]{NaOH} \underset{\text{phenyl benzoate}}{C_6H_5\text{-}COOC_6H_5} + HCl$$

d) Formation of ethers:

Williamson ether synthesis:

An alkaline solution of phenol reacts with alkyl halide to form phenyl ethers. The alkyl halide undergoes nucleophilic substitution by the phenoxide ion in the presence of alkali.

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